AAKARSH DEVULAPALLY

AI/ML Engineer | Backend Developer | NLP & LLM Integration

aakarshkashyap42@gmail.com | +91 7330855351 | LinkedIn | GitHub | Warangal, India

PROFESSIONAL SUMMARY

B.Tech (AI & ML) student at SR University with published ensemble learning research (ROC-AUC: 0.832 on 500K+ records). Skilled in building production-style NLP pipelines using Transformers and spaCy, RESTful APIs with Flask/FastAPI, and applying Bayesian hyperparameter optimization with Optuna. Actively building LLM-integrated applications, exploring RAG architectures, and developing Android/mobile apps. Strong foundation in Data Structures & Algorithms, DBMS, Operating Systems, and Computer Networks.

EDUCATION

Bachelor of Technology — Computer Science Engineering (AI & ML) **2024 – 2028 (Expected)**

SR University, Warangal, India

Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS

- Cloud Security Builder — AWS Academy
- Azure AI Fundamentals — Microsoft

TECHNICAL SKILLS

Languages	Python, JavaScript, SQL, C, Java, Kotlin
AI / ML	Scikit-learn, XGBoost, LightGBM, CatBoost, Hugging Face Transformers, spaCy, NLTK, Optuna, pandas, NumPy, Matplotlib
LLM / GenAI	OpenAI API, LangChain, Prompt Engineering, Embeddings, FAISS, RAG Pipelines, Agentic Workflows
NLP	Tokenization, Named Entity Recognition (NER), Text Summarization, Sentence Transformers, TF-IDF, Feature Engineering
Backend	Flask, FastAPI, Django, RESTful APIs, JWT Authentication, CRUD Operations, PostgreSQL, MySQL
MLOps / Tools	Git, GitHub, Docker (learning), MLflow (learning), Linux/Unix Shell Scripting
Frontend / Mobile	React, React Native, Flutter, HTML, CSS, Vite
Concepts	Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, CI/CD (basic)

RESEARCH & PUBLICATIONS

LFEKT-Based Student Performance Prediction using Ensemble Learning — *Academic Research Paper* 2024 – 2025

- Designed a hybrid knowledge tracing framework integrating Item Response Theory (IRT), Ebbinghaus forgetting curve, and temporal learning dynamics into a unified feature space of 106+ engineered features capturing student behaviour, memory decay, and knowledge progression.
- Built stacking ensemble model (XGBoost + LightGBM + CatBoost) with Bayesian hyperparameter optimization via Optuna; achieved ROC-AUC 0.832 on ASSISTments 2009–10 dataset (500K+ interaction records), outperforming LSTM baseline by +3.1%.
- Conducted ablation studies, Wilcoxon statistical significance testing, and SHAP-based feature importance analysis to validate model effectiveness and interpretability.
- Preprocessing pipeline built with pandas and NumPy; model training in Scikit-learn with custom cross-validation strategy on benchmark dataset.

PROJECTS

AI Legal Document Simplifier & Risk Analyzer — *NLP / GenAI* 2024

- Built an end-to-end NLP pipeline using spaCy and Hugging Face Transformers for legal document parsing — tokenization, named entity recognition (NER), and extractive summarization on contract-length inputs.
- Designed clause-level risk classifier (Low / Medium / High) using fine-tuned sentence embeddings; evaluated on 200 manually labelled contract clauses achieving 98.5% accuracy and 0.94 F1-score.
- Integrated OpenAI API for plain-language explanation of identified risk clauses; built prompt templates for consistent structured output.
- Built React + Vite frontend with drag-and-drop upload and interactive risk dashboard; Flask REST API backend with JWT authentication and modular architecture.

Scalable Backend API for Data Processing & Automation — *Backend* 2024

- Designed and deployed RESTful APIs using Flask and FastAPI for structured data processing, ETL pipelines, and workflow automation.
- Implemented request validation, pagination, rate limiting, and JWT-based authentication; integrated PostgreSQL for persistent storage with optimized query patterns.
- Built modular, reusable backend components following clean architecture principles; reduced manual data processing time by ~40% through automation.

Android Application Development — *Mobile Development* 2024

- Developed Android applications using Java and Kotlin with responsive UI components following Material Design principles.
- Integrated backend services (REST APIs) and local storage for efficient data management and offline capability.
- Performed systematic testing and debugging to enhance application stability and performance.

ML Classification & Regression Suite — *Academic ML* 2024

- Developed end-to-end ML pipelines in Python (Scikit-learn) for classification and regression tasks — data ingestion, preprocessing, feature engineering, model selection, and evaluation.
- Applied dimensionality reduction (PCA), feature selection (RFECV), and cross-validation strategies; reported metrics including accuracy, precision, recall, F1, and ROC-AUC.

EXPERIENCE

Software Developer (Academic) | SR University | *Warangal, India* **Aug 2024 – Present**

- Developed and maintained Python applications for data processing and workflow automation, improving task efficiency and reliability.
- Built web applications using Flask and Django — implemented backend logic, routing, database integration, and REST APIs with JWT authentication.
- Developed Android mobile applications using Java and Kotlin, implementing core features and responsive UI components.
- Participated in peer code reviews, providing constructive feedback to improve code quality, readability, and maintainability.
- Applied AI/ML concepts including data preprocessing, model development, and evaluation techniques in coursework and live projects.
- Practised Linux command-line operations and shell scripting for development and system-level tasks; collaborated using Git.

CURRENTLY BUILDING & LEARNING

- RAG-based document QA system using LangChain, FAISS, and OpenAI API — indexing chunked PDFs with semantic embeddings for retrieval-augmented generation.
- Exploring agentic workflows with LangChain Agents and tool-use patterns for multi-step AI automation tasks.
- LeetCode 100+ problems solved — Data Structures & Algorithms: Arrays, Trees, Dynamic Programming, Graphs.

References available on request